Preparation for Minitest 1 – worked solution

Logic Programming (621.315), Winter semester 2026

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Exercise 1

Given is program P. Determine all of its answer sets.

Rule numbers refer to the following order (one rule ends at each period).

```
dress :- not shirt.
shirt :- not dress.
sandals :- not sneakers.
sneakers :- not sandals.
pants :- sneakers.
skirt :- shirt, sandals.
sandals :- dress.
```

If neither `dress` nor `shirt` were in X, the reduct would contain the facts `dress` and `shirt`, because `not shirt` and `not dress` would survive. Hence exactly one of `dress`, `shirt` is in every answer set. If neither `sandals` nor `sneakers` were in X, the reduct would contain the facts `sandals` and `sneakers` for the same reason. Hence exactly one of `sandals`, `sneakers` is in every answer set. If `dress` $\in X$, then rule 7 survives in the reduct and has body `dress`, which becomes a fact. Hence `sandals` $\in \mathrm{Cn}(P^X)$, so `sandals` $\in X$. This rules out `sneakers`. The candidate is $X_1 = \{\texttt{dress}, \texttt{sandals}\}$. If `shirt` $\in X$, rule 6 survives and forces `skirt` $\in \mathrm{Cn}(P^X)$ as soon as `sandals` $\in X$. Splitting on the shoe pair gives the remaining candidates $X_2 = \{\texttt{shirt}, \texttt{sandals}, \texttt{skirt}\}$ and $X_3 = \{\texttt{shirt}, \texttt{sneakers}, \texttt{pants}\}$. These candidates exhaust the forced choices. □

Candidate X_1

$$X = \{\texttt{dress}, \texttt{sandals}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is retained because $\{\texttt{shirt}\} \cap X = \varnothing$; its default negations are removed.

Rule 2 is deleted because $\{\texttt{dress}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{sneakers}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is deleted because $\{\texttt{sandals}\} \subseteq X$.

Rule 5 is retained because it contains no default negation.

Rule 6 is retained because it contains no default negation.

Rule 7 is retained because it contains no default negation.

Rule numbers refer to the following order (one rule ends at each period).

```
dress.
sandals.
pants :- sneakers.
skirt :- shirt, sandals.
sandals :- dress.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{dress}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{sandals}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{sandals}\} \qquad \text{rule 5, head already present}$$

No rule has a true body and a head outside M_3: rule 3 needs `sneakers` and rule 4 needs `shirt`, neither of which is derivable. Hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{dress}, \texttt{sandals}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Candidate X_2

$$X = \{\texttt{shirt}, \texttt{sandals}, \texttt{skirt}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{shirt}\} \subseteq X$.

Rule 2 is retained because $\{\texttt{dress}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is retained because $\{\texttt{sneakers}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is deleted because $\{\texttt{sandals}\} \subseteq X$.

Rule 5 is retained because it contains no default negation.

Rule 6 is retained because it contains no default negation.

Rule 7 is retained because it contains no default negation.

Rule numbers refer to the following order (one rule ends at each period).

```
shirt.
sandals.
pants :- sneakers.
skirt :- shirt, sandals.
sandals :- dress.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{shirt}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{sandals}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{skirt}\} \qquad \text{rule 4}$$

No rule has a true body and a head outside M_3: rule 3 needs `sneakers` and rule 5 needs `dress`, neither of which is derivable. Hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{shirt}, \texttt{sandals}, \texttt{skirt}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Candidate X_3

$$X = \{\texttt{shirt}, \texttt{sneakers}, \texttt{pants}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{shirt}\} \subseteq X$.

Rule 2 is retained because $\{\texttt{dress}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is deleted because $\{\texttt{sneakers}\} \subseteq X$.

Rule 4 is retained because $\{\texttt{sandals}\} \cap X = \varnothing$; its default negations are removed.

Rule 5 is retained because it contains no default negation.

Rule 6 is retained because it contains no default negation.

Rule 7 is retained because it contains no default negation.

Rule numbers refer to the following order (one rule ends at each period).

```
shirt.
sneakers.
pants :- sneakers.
skirt :- shirt, sandals.
sandals :- dress.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{shirt}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{sneakers}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{pants}\} \qquad \text{rule 3}$$

No rule has a true body and a head outside M_3: rule 4 needs `sandals` and rule 5 needs `dress`, neither of which is derivable. Hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{shirt}, \texttt{sneakers}, \texttt{pants}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

The complete list of answer sets is:

$$\boxed{X_1 = \{\texttt{dress}, \texttt{sandals}\}}$$

$$\boxed{X_2 = \{\texttt{shirt}, \texttt{sandals}, \texttt{skirt}\}}$$

$$\boxed{X_3 = \{\texttt{shirt}, \texttt{sneakers}, \texttt{pants}\}}$$

Exercise 2

Given is program P. Determine whether the following sets are stable models: {b, c, d, h}; {a, f}; {b, d, e, g, h}.

Rule numbers refer to the following order (one rule ends at each period).

```
a :- not b, not d.
b :- not a.
c :- b, not e.
d :- c.
e :- d, not c.
f :- a, not g.
g :- not f.
h :- b, g.
```

Candidate X_1

$$X = \{\texttt{b},\ \texttt{c},\ \texttt{d},\ \texttt{h}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{b},\ \texttt{d}\} \subseteq X$.

Rule 2 is retained because $\{\texttt{a}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is retained because it contains no default negation.

Rule 5 is deleted because $\{\texttt{c}\} \subseteq X$.

Rule 6 is retained because $\{\texttt{g}\} \cap X = \varnothing$; its default negations are removed.

Rule 7 is retained because $\{\texttt{f}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is retained because it contains no default negation.

Rule numbers refer to the following order (one rule ends at each period).

```
b.
c :- b.
d :- c.
f :- a.
g.
h :- b, g.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 2}$$

$$M_2 = M_1 \cup \{\texttt{g}\} \qquad \text{fact, rule 7}$$

$$M_3 = M_2 \cup \{\texttt{c}\} \qquad \text{rule 3}$$

$$M_4 = M_3 \cup \{\texttt{d}\} \qquad \text{rule 4}$$

$$M_5 = M_4 \cup \{\texttt{h}\} \qquad \text{rule 8}$$

No rule has a true body and a head outside M_5: rule 6 needs a, which is not derivable. Hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b},\ \texttt{c},\ \texttt{d},\ \texttt{g},\ \texttt{h}\}$$

$$\mathrm{Cn}(P^X) \neq X$$

$$\boxed{X \text{ is not an answer set.}}$$

The atom g is in $\mathrm{Cn}(P^X)$ but missing from the candidate.

Candidate X_2

$$X = \{\texttt{a}, \texttt{f}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is retained because $\{\texttt{b}, \texttt{d}\} \cap X = \varnothing$; its default negations are removed.

Rule 2 is deleted because $\{\texttt{a}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is retained because it contains no default negation.

Rule 5 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is retained because $\{\texttt{g}\} \cap X = \varnothing$; its default negations are removed.

Rule 7 is deleted because $\{\texttt{f}\} \subseteq X$.

Rule 8 is retained because it contains no default negation.

Rule numbers refer to the following order (one rule ends at each period).

```
a.
c :- b.
d :- c.
e :- d.
f :- a.
h :- b, g.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{a}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{f}\} \qquad \text{rule 6}$$

No rule has a true body and a head outside M_2: rule 3 needs `b` and rule 5 needs `d`, neither of which is derivable. Hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a}, \texttt{f}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Candidate X_3

$$X = \{\texttt{b}, \texttt{d}, \texttt{e}, \texttt{g}, \texttt{h}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{b}, \texttt{d}\} \subseteq X$.

Rule 2 is retained because $\{\texttt{a}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is deleted because $\{\texttt{e}\} \subseteq X$.

Rule 4 is retained because it contains no default negation.

Rule 5 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is deleted because $\{\texttt{g}\} \subseteq X$.

Rule 7 is retained because $\{\texttt{f}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is retained because it contains no default negation.

Rule numbers refer to the following order (one rule ends at each period).

```
b.
d :- c.
e :- d.
g.
h :- b, g.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 2}$$

$$M_2 = M_1 \cup \{\texttt{g}\} \qquad \text{fact, rule 7}$$

$$M_3 = M_2 \cup \{\texttt{h}\} \qquad \text{rule 8}$$

No rule has a true body and a head outside M_3: rule 3 needs `c`, which is not derivable. Hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b}, \texttt{g}, \texttt{h}\}$$

$$\mathrm{Cn}(P^X) \neq X$$

$$\boxed{X \text{ is not an answer set.}}$$

The atoms `d` and `e` are in the candidate but not in $\mathrm{Cn}(P^X)$.